Day 130 | Unit 09 | Teacher plan

Display, Describe, and Summarize Data

Critique Statistical Summaries

Standards: 6.SP.B.5d, 6.SP.A.2, 6.SP.A.3

Learning target: I can critique Statistical Summaries, and explain and check my reasoning.

enVision Topic 8: Display, Describe, and Summarize Data | Spectrum companion Chapter 7

0–5 minutes / Launch

Read the target. Ask for an estimate or a sketch before calculating. Connect the new idea to the previous lesson.

5–17 / Model

Use Worked Example 1, then Example 2. Ask students to name the next step and explain a choice. Use the shape and context. Mean with mean absolute deviation can describe fairly symmetric data. Median with interquartile range often describes skewed data or data with an unusual value more fairly. A distribution shows which values occur and how often. Describe a typical value, how spread out the values are, and whether the shape is roughly symmetric, skewed, clustered, or has unusual values.

17–42 / Practice

Assign questions 1–10 in order. Use 1–3 for foundations, 4–7 for application, and 8–10 for state-format response and reasoning. The example button provides a separate question with a worked method.

42–50 / Discuss and check

Compare representations. Review a selected answer and a plausible error. Use question 10 as an exit discussion; ask students to check units, scale, or substitution.

Support and extension

Offer number lines, fraction strips, grid paper, or a table template as appropriate. Read multi-part directions one part at a time. For extension, ask for another representation or a counterexample.

Companion and state alignment

Use the matching skill in the edition you own. No unverified publisher page numbers are assigned. AASA references below identify related public sample items by standard and format; all course questions are original.

AASA sample item references: 10

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01 / Learn

Use the shape and context. Mean with mean absolute deviation can describe fairly symmetric data. Median with interquartile range often describes skewed data or data with an unusual value more fairly. A distribution shows which values occur and how often. Describe a typical value, how spread out the values are, and whether the shape is roughly symmetric, skewed, clustered, or has unusual values.

Worked example 1

Reading times in minutes are 3, 4, 4, 5, 5, 6. Choose a center and matching variability measure. Calculate both and justify your choice.

1. Use the shape and context.

2. Mean with mean absolute deviation can describe fairly symmetric data.

3. Mean = 9/2, median = 9/2, IQR = 5 - 4 = 1, MAD = 5/6.

4. Accept another justified choice tied to shape and purpose.

5. State the result in the form requested: Mean 9/2 min and MAD 5/6 min; roughly symmetric data.

Answer: Mean 9/2 min and MAD 5/6 min; roughly symmetric data

Worked example 2

Six students recorded reading times (minutes): 6, 12, 12, 18, 18, 24. Part A: describe center and spread using appropriate measures. Part B: describe shape, count, units, and any striking value; justify the chosen measures.

1. A distribution shows which values occur and how often.

2. Order the observations.

3. Middle positions give median 15 and half medians give IQR 6.

4. Mean is 90/6

5. averaging distances gives MAD 5.

6. Tie the chosen summary to the displayed shape and context.

Answer: 6 observations; minutes; median 15, IQR 6; mean 15, MAD 5. Roughly balanced; mean/MAD or median/IQR with justification.

Day 130 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Reading times in minutes are 7, 8, 8, 9, 9, 10. Choose a center and matching variability measure. Calculate both and justify your choice.

02 / Constructed response

Reading times in minutes are 8, 9, 9, 10, 10, 11. Describe a typical value, range, shape, and any striking value.

03 / Constructed response

Reading times in minutes are 6, 7, 7, 8, 8, 9. Choose a center and matching variability measure. Calculate both and justify your choice.

04 / Constructed response

Reading times in minutes are 2, 6, 6, 10, 10, 14. Describe a typical value, range, shape, and any striking value.

05 / Constructed response

Reading times in minutes are 7, 11, 11, 15, 15, 19. Choose a center and matching variability measure. Calculate both and justify your choice.

06 / Constructed response

Reading times in minutes are 3, 8, 8, 13, 13, 18. Describe a typical value, range, shape, and any striking value.

07 / Constructed response

Reading times in minutes are 6, 7, 7, 8, 9, 18. Choose a center and matching variability measure. Calculate both and justify your choice.

08 / Statistical reasoning and explanation

Six students recorded reading times (minutes): 3, 5, 5, 7, 7, 9. Part A: describe center and spread using appropriate measures. Part B: describe shape, count, units, and any striking value; justify the chosen measures.

09 / Statistical reasoning and explanation

Six students recorded reading times (minutes): 4, 8, 8, 12, 12, 24. Part A: describe center and spread using appropriate measures. Part B: describe shape, count, units, and any striking value; justify the chosen measures.

10 / Constructed response

Reading times in minutes are 8, 9, 9, 10, 11, 20. Describe a typical value, range, shape, and any striking value.

Day 130 | Unit 09 | Answer key

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Learning target: I can critique Statistical Summaries, and explain and check my reasoning.

1. Mean 17/2 min and MAD 5/6 min; roughly symmetric data

Mean = 17/2, median = 17/2, IQR = 9 - 8 = 1, MAD = 5/6. Accept another justified choice tied to shape and purpose.

6.SP.B.5d | Constructed response

2. Center near 19/2 min; range 3 min; roughly symmetric; no isolated extreme

The values balance around the middle. Range = 11 - 8.

6.SP.A.2, 6.SP.A.3 | Constructed response

3. Mean 15/2 min and MAD 5/6 min; roughly symmetric data

Mean = 15/2, median = 15/2, IQR = 8 - 7 = 1, MAD = 5/6. Accept another justified choice tied to shape and purpose.

6.SP.B.5d | Constructed response

4. Center near 8 min; range 12 min; roughly symmetric; no isolated extreme

The values balance around the middle. Range = 14 - 2.

6.SP.A.2, 6.SP.A.3 | Constructed response

5. Mean 13 min and MAD 10/3 min; roughly symmetric data

Mean = 13, median = 13, IQR = 15 - 11 = 4, MAD = 10/3. Accept another justified choice tied to shape and purpose.

6.SP.B.5d | Constructed response

6. Center near 21/2 min; range 15 min; roughly symmetric; no isolated extreme

The values balance around the middle. Range = 18 - 3.

6.SP.A.2, 6.SP.A.3 | Constructed response

7. Median 15/2 min and IQR 2 min; high value makes these resistant measures useful

Mean = 55/6, median = 15/2, IQR = 9 - 7 = 2, MAD = 53/18. Accept another justified choice tied to shape and purpose.

6.SP.B.5d | Constructed response

8. 6 observations; minutes; median 6, IQR 2; mean 6, MAD 5/3. Roughly balanced; mean/MAD or median/IQR with justification.

Order the observations. Middle positions give median 6 and half medians give IQR 2. Mean is 36/6; averaging distances gives MAD 5/3. Tie the chosen summary to the displayed shape and context.

6.SP.A.2, 6.SP.A.3 | Statistical reasoning and explanation

9. 6 observations; minutes; median 10, IQR 4; mean 34/3, MAD 14/3. A high value makes median/IQR a useful choice.

Order the observations. Middle positions give median 10 and half medians give IQR 4. Mean is 68/6; averaging distances gives MAD 14/3. Tie the chosen summary to the displayed shape and context.

6.SP.B.5d | Statistical reasoning and explanation

10. Center near 19/2 min; range 12 min; right-skewed with high value 20 min

The values cluster low with a long high tail. Range = 20 - 8.

6.SP.A.2, 6.SP.A.3 | Constructed response